hdoc day 44 by trishil chauhan

Aim

Write a menu-driven C program using separate user-defined functions to reverse each word in a sentence without changing word order, and to print the initials of a name.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 500

void readLine(char str[], int size) {
    if (fgets(str, size, stdin) != NULL) {
        str[strcspn(str, "\n")] = '\0';
    }
}

void reverseEachWord(char str[]) {
    int start = 0, i, end;
    int len = (int)strlen(str);

    for (i = 0; i <= len; i++) {
        if (isspace((unsigned char)str[i]) || str[i] == '\0') {
            end = i - 1;
            while (start < end) {
                char temp = str[start];
                str[start] = str[end];
                str[end] = temp;
                start++;
                end--;
            }
            start = i + 1;
        }
    }
}

void printInitials(const char name[]) {
    int i, newWord = 1, found = 0;
    printf("Initials: ");
    for (i = 0; name[i] != '\0'; i++) {
        if (isspace((unsigned char)name[i])) {
            newWord = 1;
        } else if (newWord) {
            printf("%c.", toupper((unsigned char)name[i]));
            newWord = 0;
            found = 1;
        }
    }
    if (!found) printf("(no name entered)");
    printf("\n");
}

int main(void) {
    char text[MAX];
    char choiceLine[32];
    int choice;
```

```
    do {
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        readLine(choiceLine, sizeof(choiceLine));
        if (sscanf(choiceLine, "%d", &choice) != 1) {
            choice = 0;
        }

        switch (choice) {
            case 1:
                printf("Enter a sentence: ");
                readLine(text, MAX);
                reverseEachWord(text);
                printf("Result: %s\n", text);
                break;
            case 2:
                printf("Enter full name: ");
                readLine(text, MAX);
                printInitials(text);
                break;
            case 3:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice. Try again.\n");
        }
    } while (choice != 3);

    return 0;
}
```

Sample Output

```
--- STRING OPERATIONS MENU ---
1. Reverse each word in a sentence
2. Print initials of a name
3. Exit
Enter your choice: 1
Enter a sentence: What is special?
Result: tahW si ?laiceps

Enter your choice: 2
Enter full name: Alexander Grahambell
Initials: A.G.

Enter your choice: 3
Exiting program.
```

Explanation

The first function reverses characters within each whitespace-separated word while preserving the original word order and spaces. The second function prints the uppercase first letter of each name component followed by a period. The program repeats the menu until Exit is selected.

Note: Character-by-character reversal of "What" produces "tahW" (preserving the uppercase W). The example "tahw" assumes lowercase output; the program preserves the input letter case.